Landmark Detection for Cephalometric X-Ray Images

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Abstract

In recent years, there has been growing interest in developing automated methods for cephalometric landmark detection using machine learning techniques. In this paper, we propose a novel approach using deep-learning-based methods to accurately and efficiently detect cephalometric landmarks. Specifically, we utilize convolutional neural networks with ResNet and U-Net architectures, which have demonstrated superior performance in various computer vision tasks. Our models are trained on an augmented dataset of cephalometric X-ray images, with horizontal and vertical flips followed by three different rotations, allowing them to learn highly discriminative features that are crucial for landmark detection. We evaluate the performance of our models using standard evaluation metrics and demonstrate that they achieve considerable precision in detecting cephalometric landmarks. However, we also identify areas for improvement, particularly in terms of pinpoint detection, and highlight the potential for further development in this area. The proposed methods have the potential to significantly improve the accuracy and efficiency of orthodontic diagnosis and treatment planning. Accurate cephalometric landmark detection is essential for the diagnosis of various orthodontic conditions, and automated methods can help reduce the time and effort required for manual detection. By improving the efficiency and accuracy of diagnosis and treatment planning, our proposed methods can ultimately lead to better patient outcomes.

1. Introduction

Cephalometric landmark detection is an extensively studied area in the fields of computer vision and medical imaging. It is a task that employs the technique of keypoint annotation, which involves identifying critical landmarks on objects within images. In this context, it entails labeling distinct anatomical points in the craniofacial area, such as the nose tip and jawbone contours. This process is a crucial preliminary step for identifying patients with skeletal and dental abnormalities. These marked points can further be utilized to aid in diagnosing and providing treatment plans for various medical conditions.

No.	Anatomical Landmarks
1	sella turcica
2	nasion
3	orbitale
4	porion
5	subspinale
6	supramentale
7	pogonion
8	menton
9	gnathion
10	gonion
11	lower incisal incision
12	upper incisal incision
13	upper lip
14	lower lip
15	subnasale
16	soft tissue pogonion
17	posterior nasal spine
18	anterior nasal spine
19	articulate

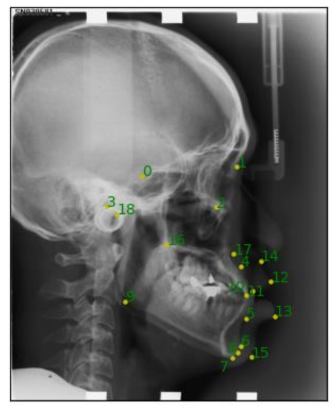

Figure 1. Name and location of cephalometric landmarks

However, manually annotating landmarks is a time-consuming, labor-intensive process that is susceptible to human errors. Current landmark detection approaches encompass both regression-based methods and deep-learning-based models [4]. Our team proposes convolutional neural networks and an encoder-decoder structure to enhance the cephalometric landmark detection process while maintaining high accuracy.

2. Related Work

Previous research in keypoint detection has often begun with segmentation, as seen in human posture or facial keypoint identification, such as in [5, 3]. This process typically involves creating a heatmap from an image and subsequently using it to identify the keypoints.

In the context of cephalometric landmark detection, there are two main approaches. The first involves utilizing machine learning models like random forests, as evidenced in [1, 2]. The second approach employs deep learning models that harness the power of convolutional neural networks or transformers, as showcased in [6, 7]. Due to the immense scale of predicting all 19 landmarks simultaneously, the paper [6] initially partitions the image into smaller patches,

Model	Parameters	In-Channel
CNN	1081064834	1
Deeper CNN	107981786	1
U-NET	31032998	1 or 3
ResNet50	24699750	3

Table 1. Experimental results. We record the number of parameters in the models. Note that ResNet50 only takes input image with 3 channels.

each containing only one landmark. Following this, convolutional neural network models are applied to each of the 19 landmarks.

3. Materials and Models

The dataset comes from 2015 ISBI Grand Challenge and comprises 150 training images, 150 validation images, and 100 testing images[1]. Each image has dimensions of 1935×2400 pixels and is in TIFF format. Two experts provided annotations for the coordinates, with the senior annotation (as opposed to the junior) being used. Each image features 19 landmarks, as depicted in Figure 1.

The model we propose is illustrated in Figure 2 with three components: pre-processing, training, and predicting.

For data pre-processing, we resize the image to 224×224 for all 400 images. Five different augmentations are then applied to the training dataset: horizontal and vertical flips followed by three different rotations (90°, 180°, and 270°). After the augmentation, we have 900 images for training. The corresponding coordinates are adjusted as well and stored in float data type.

In the training step, we experiment with four different models: CNN, a deeper CNN variant, ResNet, and U-NET. The starting code we implemented comes from GitHub[2]. A more detailed information of those models can be found in Figure 3 and Table 1. The CNN model has a basic architecture with 3 stacks of convolutional, ReLU, and pooling layers followed by fully connected layers. The deeper CNN variant has an additional convolutional layer compared to the basic CNN model.

The ResNet50 model is based on the ResNet50 architecture, which has 50 layers, including convolutional, batch normalization, ReLU, and fully connected layers. We use a pre-trained ResNet50 with approximately 25.6 million parameters, and customize the fully connected layers with an output size equal to 38, which corresponds to the number of landmarks. Since the ResNet50 model cannot take an input channel equal to 1 for grayscale images, we prepare a separate dataloader for this model and set the channel to be 3 as RGB images.

The U-Net model is designed from scratch with an encoder part consisting of four sets of double convolution blocks with ReLU activation, followed by a max pooling layer. The decoder part consists of four sets of up-convolution blocks. The final layers include an adaptive average pooling layer followed by a 1×1 convolution layer to generate output with size 38. The skip connections between layers in the encoder and decoder paths help the network better localize and preserve high-resolution features.

In the predicting step, we evaluate the models on the test set, which consists of 100 images. We use the mean absolute error (MAE) as the evaluation metric to compare the predicted landmarks with the ground truth landmarks. We employ tools such as PyTorch, PyTorch Lightning, and Weights & Biases[3] for implementing the models and evaluating the results.

4. Experiments and Results

Unless otherwise mentioned, the hyperparameters we set batch size to 10, learning rate to $1e - 3$. We performed different optimizers, batch sizes, learning rates, etc.

Mean square error (MSE) and Dice loss are two common loss functions used in keypoint detection. In our implementation, we employ the mean square error (MSE) as the loss function. To calculate the accuracy, we set a threshold value (10). If the MSE between a specific landmark prediction and its corresponding ground truth coordinates is less than this threshold, we consider it a correct prediction; otherwise, it is regarded as a false prediction.

In Figure 4 and Figure 5, we can compare both the training loss and validation loss of the four models.

A sample prediction using our U-Net model is shown in Figure 7. However, it is important to note that pinpointing landmarks on X-ray images is a challenging task, especially for certain landmarks, such as those located in the soft tissue regions. To account for this variability in the data and ensure that our model can detect landmarks with a certain level of precision, we deliberately set a small threshold in the accuracy determination for this task. As a result, the accuracy achieved may not be as desirable as we would like, but it is still a significant improvement over manual detection methods.

We also notice that the difference between using a pre-trained ResNet-50 and using a self-trained ResNet 50 is minimal. Our hypothesis is that the ResNet-50 is pre-trained on the ImageNet, which does not share many similarities with medical images like our dataset. The accuracy of prediction for the pre-trained ResNet-50 is shown in Figure 6.

[1] The official website is no longer maintained, but the data can be found on figshare or Kaggle: https://figshare.com/s/37ec464af8e81ae6ebbf

[2] https://github.com/fronay/ceph_xray_keypoints

[3] https://wandb.ai/home

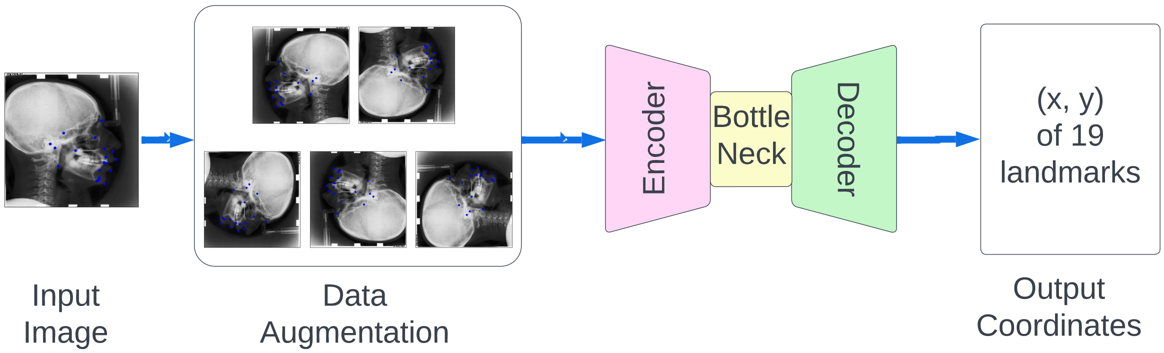

Figure 2. This figure illustrates the components of our experiments. The input is an image and it is augmented via flipping and rotating. The output is an array of 2-dimensional coordinates for the 19 landmarks.

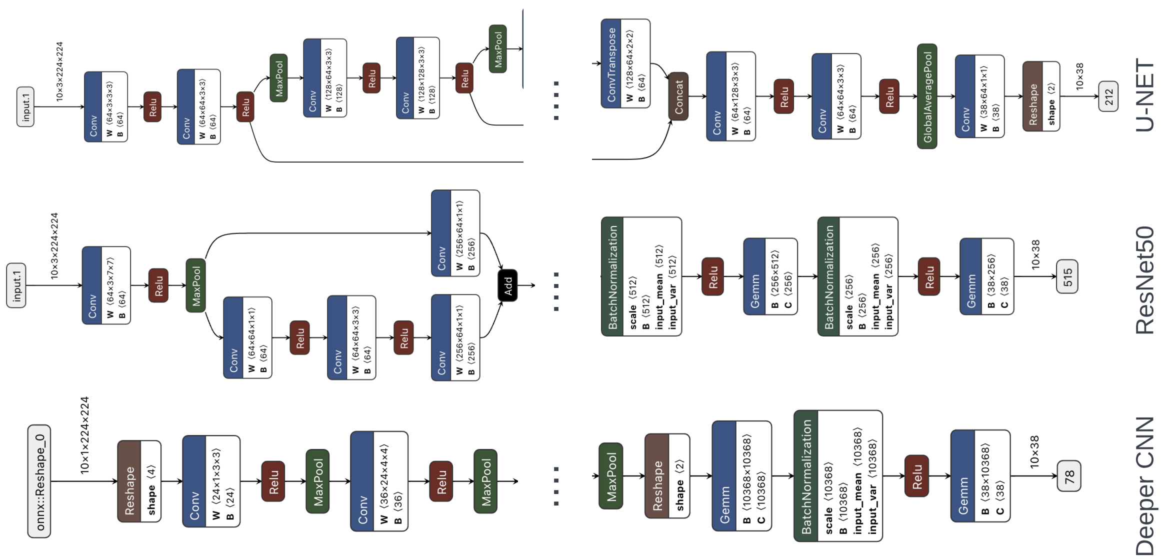

Figure 3. Model Details

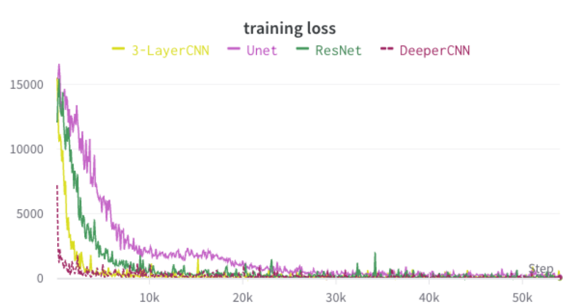

Figure 4. Training loss of 4 models

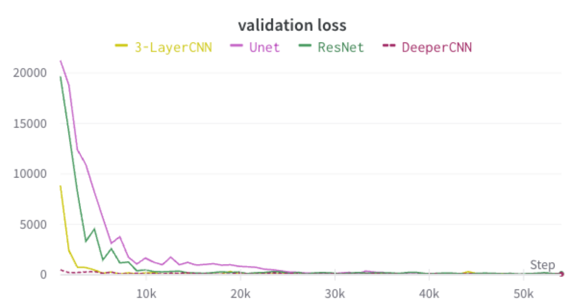

Figure 5. Validation loss of 4 models

To improve our model's performance, it may be worthwhile to explore alternative approaches, such as predicting landmarks sequentially, one point at a time. This could potentially enhance accuracy and precision, as it would allow

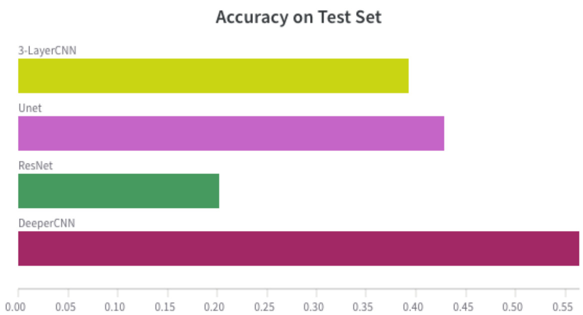

Figure 6. Validation loss of 4 models

the model to focus on individual landmarks in isolation, rather than predicting them all at once. This approach could also potentially reduce the impact of errors on nearby landmarks, leading to more accurate and precise results.

In future work, we also plan to further investigate the impact of different data augmentation techniques on our model's performance, as well as exploring different network architectures and hyperparameter configurations. Overall, we believe that our proposed deep-learning-based method for cephalometric landmark detection has significant potential for improving the accuracy and efficiency of orthodontic diagnosis and treatment planning.

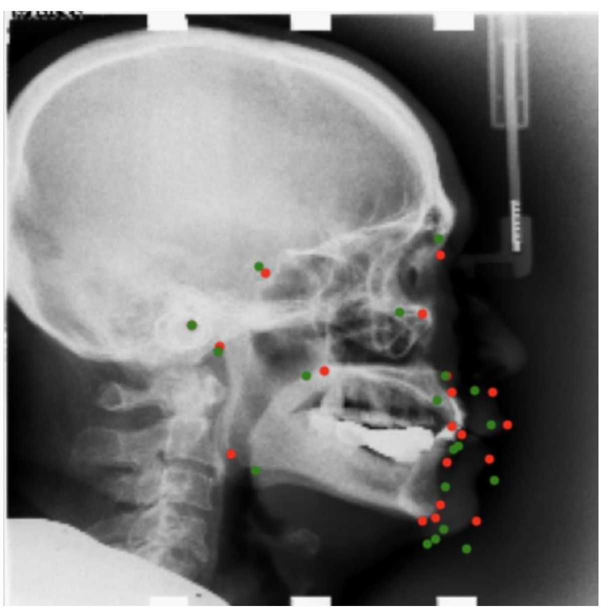

Figure 7. A sample of prediction using U-Net model. The predicted points are in green and ground truth landmarks are in red.

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